

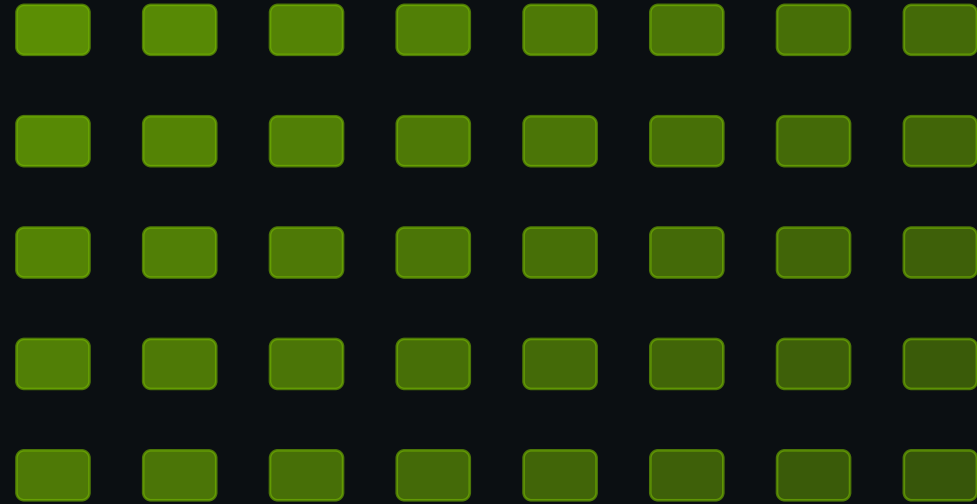
Seminar 2

CUDA Accelerated Hard NMS Baseline · V1 · V2 with Numba

Class-aware greedy NMS: from CPU reference to GPU
pairwise acceleration

Team: Tuấn · Tân

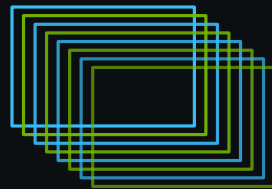
Course: Applied Parallel Programming



CPU baseline → V1 full IoU → V2 packed mask →
CUDA evidence gate

Problem Motivation

Object detectors produce many overlapping boxes



NMS removes redundant detections while keeping the highest-confidence box.

CPU Greedy NMS

Baseline algorithm and complexity



Sorting: $O(N \log N)$

Greedy suppression: $\sim O(N^2)$

IoU repeated many times

CPU Baseline Architecture

End-to-end reference implementation



WHY THIS MATTERS

CPU baseline is the greedy hard-NMS reference.
Profiling identifies pairwise IoU as the target.
V1/V2 must reproduce this output.

CPU Profiling: Pairwise IoU Is the Target

Local CPU · N = 10,000 · cProfile evidence

18,313

function calls

0.449s

total runtime

6,100

IoU calls

40.3%

IoU share

0.001s

argsort

`run_cpu()`



0.449s cumulative

`iou_one_to_many()`



0.181s total

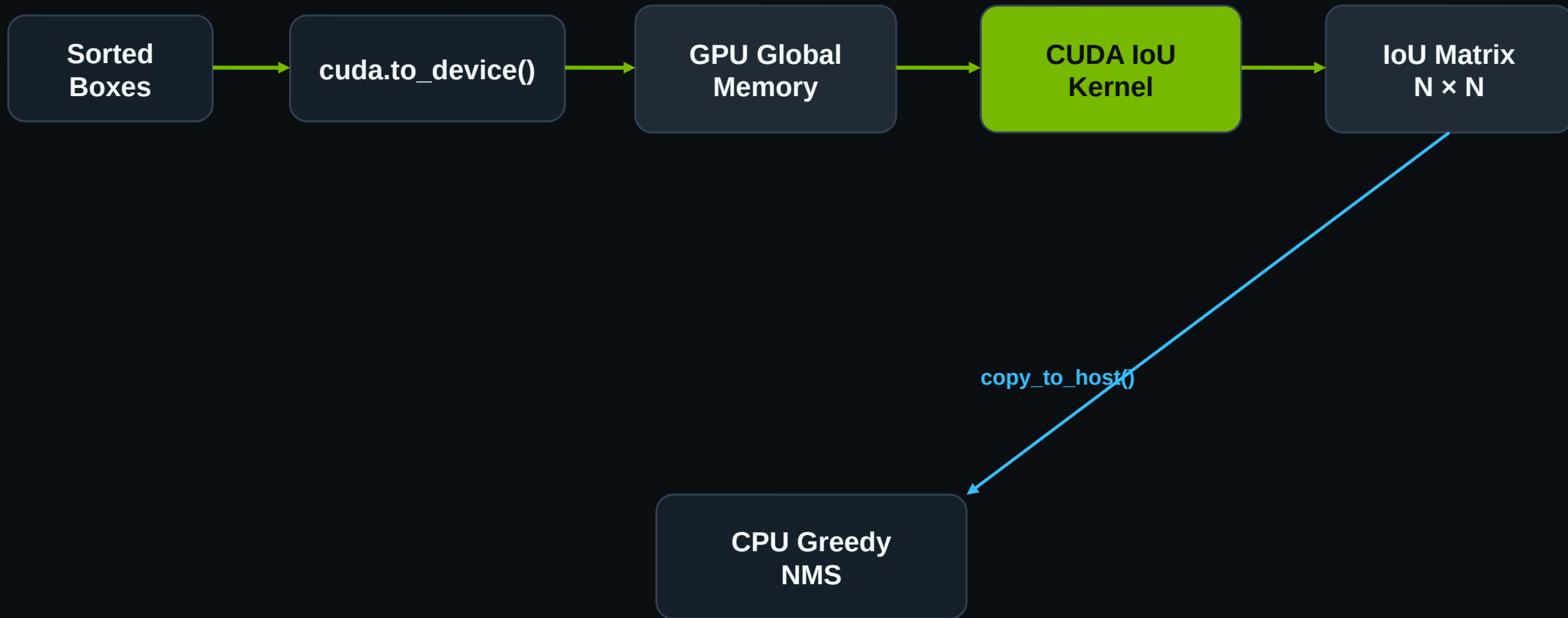
`argsort()`



0.001s cumulative

GPU Version 1 Architecture

Parallel IoU matrix, CPU greedy suppression



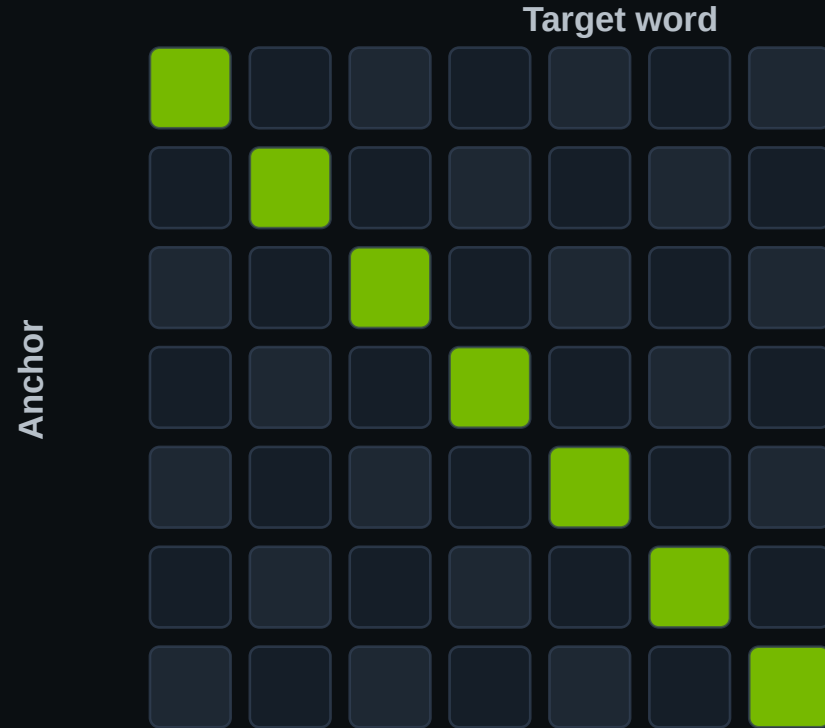
GPU does the $O(N^2)$ pairwise work

CPU keeps simple greedy control flow

V1 is easy to test and explain

GPU V2: Batched Packed Suppression Mask

grid = (anchor word, target word, image)



Thread(anchor)
→
one uint64 mask word

$\text{blockIdx.x} = \text{anchor} \cdot y = \text{target} \cdot z = \text{image}$

SoA layout enables coalesced coordinate reads.
Each block caches 64 target boxes in shared memory.
One bit records whether IoU exceeds the threshold.

Correctness & Environment

Local correctness is verified; CUDA parity is the next evidence gate



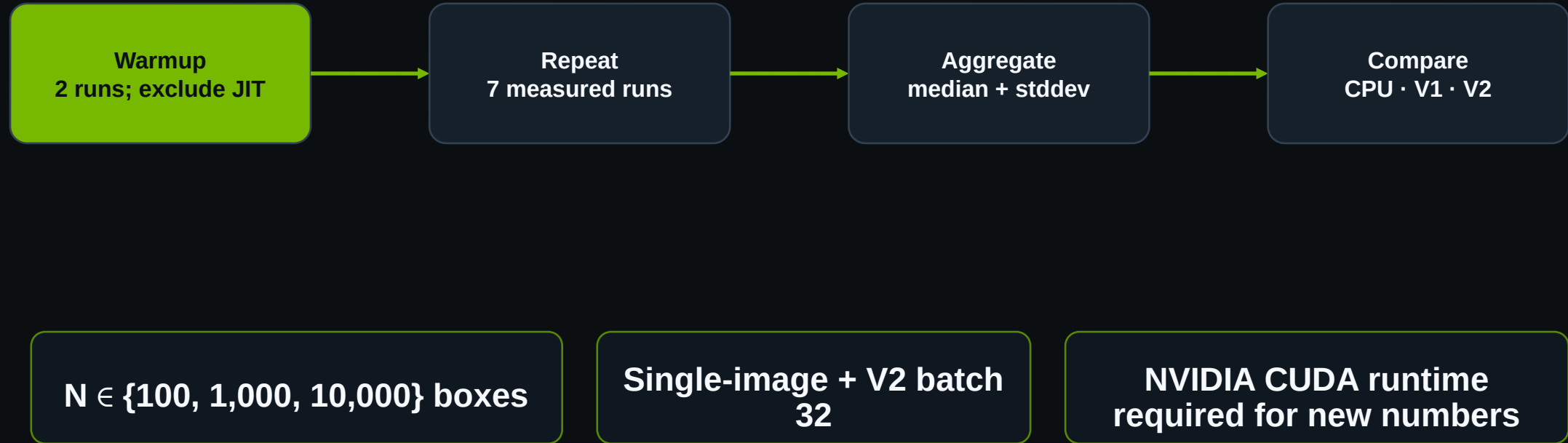
macOS · no NVIDIA CUDA

Python 3.12.13 · NumPy 2.0.2

Hard-NMS scope: baseline / V1 / V2

Benchmark protocol for the next CUDA rerun

Repeatable end-to-end latency after CUDA warmup



Batch benchmark: required CUDA evidence

Candidate protocol: batch $32 \times 10,000$ boxes · end to end

CURRENT STATUS

Local verification: 30 passed, 41 CUDA-skipped
Batch input: 32 images $\times$ 10,000 boxes
Current CUDA benchmark: pending rerun
No current CUDA number is reported

CATALOG TARGET & EVIDENCE GAP



$O(B \cdot N^2)$ pairwise work

Packed mask copied to host · greedy resolution
on CPU

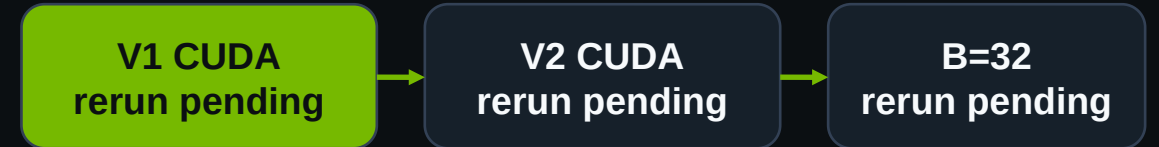
Limitations / Honest Conclusion

Evidence gap and serial greedy dependency remain

CURRENT LIMITATIONS

V1 transfers the full IoU matrix GPU → CPU.
V2 packs uint64 masks, but final greedy decisions stay on CPU.
This dependency is the remaining hard-NMS limit.

WHAT IS STILL REQUIRED



CUDA correctness and timing must be rerun

Greedy resolution stays on CPU

References & Reproducibility

Code, correctness evidence, and measured results

Applied Parallel Programming — Track A4: Real-Time Non-Maximum Suppression.

Current modules: baseline/, v1/, v2/ and compatibility entry points.

Greedy reference: CPU baseline and per-class torchvision.ops.nms.

The seminar scope is baseline, v1, and v2 hard-NMS.

New CUDA evidence must be produced on the commit being presented.

Thank you